

PLANT LEAF DISEASE REGION SEGMENTATION AND SEVERITY ANALYSIS USING DIGITAL IMAGE PROCESSING

Digital Image Processing Mini Project

Submitted By,

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1. Title

Plant Leaf Disease Region Segmentation and Severity Analysis Using Digital Image Processing

2. Problem Statement

Plant diseases can affect the health and growth of plants. Identifying the affected parts of a leaf can help in understanding the level of disease.

In this project, we use Digital Image Processing techniques to identify and segment the diseased regions in plant leaf images. The project focuses on finding the affected regions rather than classifying the type of disease.

Different segmentation techniques are applied and compared to see how well they identify the disease regions. The affected area is also estimated to give a simple severity indication.

3. Objectives

The main objectives of this project are:

- To work with plant leaf images and their corresponding disease masks.
- To preprocess the leaf images using grayscale conversion and Gaussian blurring.
- To segment disease regions using Otsu Thresholding, Adaptive Thresholding, and Canny Edge Detection.
- To improve the segmentation output using morphological operations.
- To compare the segmentation methods using IoU, Dice Coefficient, and Accuracy.
- To estimate the affected area of the leaf.
- To provide a simple severity level based on the estimated affected area.
- To analyze the limitations of the segmentation results.

4. Dataset

For this project, we used a plant leaf disease segmentation dataset containing leaf images and their corresponding ground truth masks.

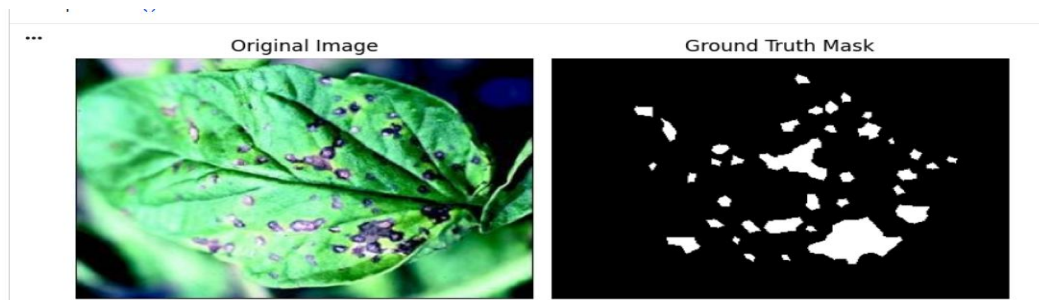
The dataset contains **2,940 leaf images** and **2,940 mask images** in the augmented dataset used for this project. Each image has a corresponding mask that represents the disease-affected region.

The images are in colour format, while the masks are grayscale images. The first sample image used in the project has a size of **235 × 326 pixels**.

The dataset was used to test different image segmentation techniques and to compare the predicted disease regions with the ground truth masks.

Dataset Structure

- **Images:** 2,940
- **Masks:** 2,940
- **Image type:** Colour images
- **Mask type:** Grayscale masks
- **Image size:** 235 × 326 pixels for the sample image
- **Ground truth:** Used to evaluate the segmentation results



Dataset Limitation

The dataset contains a fixed set of leaf images, so the performance of the techniques may change when they are applied to leaves with different lighting conditions, backgrounds, colours, or disease patterns.

```
1] Os images_path = "/content/leaf_dataset/aug_data/aug_data/images"
    masks_path = "/content/leaf_dataset/aug_data/aug_data/masks"

    image_files = sorted(os.listdir(images_path))
    mask_files = sorted(os.listdir(masks_path))

    print("Images:", len(image_files))
    print("Masks:", len(mask_files))

Images: 2940
Masks: 2940

1] Os image = cv2.imread(os.path.join(images_path, image_files[0]))
    mask = cv2.imread(os.path.join(masks_path, mask_files[0]), cv2.IMREAD_GRAYSCALE)

    print("Image shape:", image.shape)
    print("Mask shape:", mask.shape)

Image shape: (235, 326, 3)
Mask shape: (235, 326)
```

5. Methodology

The project follows a simple image processing workflow to identify the disease-affected regions in plant leaves. First, the input image is converted into grayscale and smoothed using Gaussian Blur. Then, different segmentation techniques are applied to identify the affected regions. The methods used are Otsu Thresholding, Adaptive Thresholding, and Canny Edge Detection.

The segmentation results are compared with the ground truth masks using evaluation measures such as IoU, Dice Coefficient, and Accuracy. Finally, the affected area is estimated using the selected segmentation result.

5.1 Preprocessing

Before applying segmentation techniques, the input leaf image is preprocessed to make it easier to analyze.

First, the colour image is converted into a **grayscale image**. This reduces the image to intensity values and makes the segmentation process simpler.

After grayscale conversion, **Gaussian Blur** is applied using a 5×5 kernel. This helps reduce small noise and smooth the image before thresholding and edge detection.

The preprocessing steps used are:

1. Convert the colour image to grayscale.
2. Apply Gaussian Blur using a 5×5 kernel.
3. Use the processed image for segmentation.

5.2 Otsu Thresholding

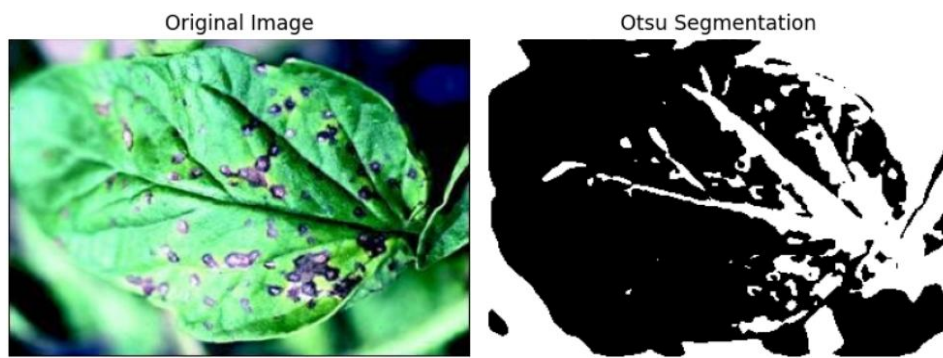
Otsu Thresholding is used to separate the image into different regions based on pixel intensity. In this project, the blurred grayscale image is used as the input.

Otsu's method automatically selects a threshold value from the image instead of requiring us to choose one manually. The selected threshold is then used to create a binary mask of the possible affected regions.

After thresholding, the segmentation result is displayed and compared with the ground truth mask using IoU, Dice Coefficient, and Accuracy.

For the sample image, the Otsu threshold value obtained was **117**.

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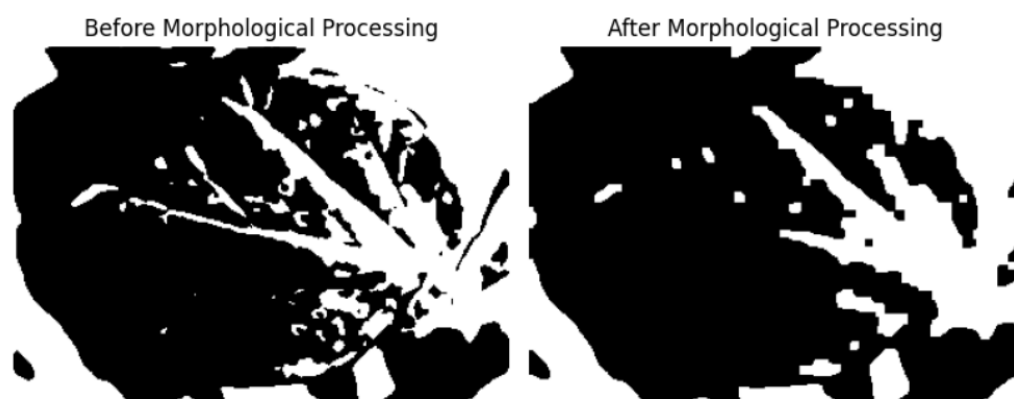


5.3 Morphological Processing

After Otsu thresholding, some small unwanted regions or gaps may be present in the segmentation mask. To improve the mask, morphological operations are applied.

A 5×5 kernel is used for the processing. First, **morphological opening** is applied to remove small unwanted regions. Then, **morphological closing** is applied to fill small gaps and make the segmented regions more connected.

This step helps produce a cleaner segmentation mask before evaluating the result.



5.4 Adaptive Thresholding

Adaptive Thresholding is used to segment the image when the lighting or intensity is not uniform across the leaf. Unlike a single global threshold, it calculates the threshold value for different local regions of the image.

In this project, Gaussian Blur is applied before Adaptive Thresholding. The Gaussian adaptive method is used with a block size of 11 and a constant value of 2.

The resulting binary image represents the possible disease-affected regions. This method is useful for handling images where different parts of the leaf have different intensity levels.

...

Original Image



Adaptive Thresholding



5.5 Canny Edge Detection

Canny Edge Detection is used to detect important boundaries or edges in the leaf image. It helps identify areas where there is a significant change in pixel intensity.

In this project, Canny Edge Detection is applied to the blurred grayscale image using threshold values of 50 and 150. The detected edges are then converted into a filled binary mask to represent the possible affected regions.

This method is included in the project to compare an edge-based technique with the threshold-based segmentation methods.

...

Original Image



Canny Edge Detection



5.6 Evaluation and Comparison

The segmentation methods were evaluated by comparing their predicted masks with the ground truth masks.

Three methods were compared:

- Otsu Thresholding
- Adaptive Thresholding

- Canny Edge Detection

The comparison was performed on **50 leaf images** using Average IoU (Intersection over Union). IoU measures how much the predicted region overlaps with the actual disease region.

The obtained Average IoU values are:

Segmentation Method	Average IoU
Otsu Thresholding	0.0711
Adaptive Thresholding	0.1547
Canny Edge Detection	0.1419

From the comparison, **Adaptive Thresholding achieved the highest Average IoU of 0.1547** among the three methods tested. Therefore, Adaptive Thresholding was selected as the final segmentation method for this project.

The selected method was further evaluated using Dice Coefficient and Accuracy. The average values obtained were:

- **Average Dice Coefficient:** 0.2565
- **Average Accuracy:** 0.7388

These results show that Adaptive Thresholding performed better than Otsu Thresholding and Canny Edge Detection in terms of average IoU for the tested images.

```
... Average IoU - Otsu: 0.0711
    Average IoU - Adaptive: 0.1547
    Average IoU - Canny: 0.1419
```

```
Average Dice: 0.2565
Average Accuracy: 0.7388
```

5.7 Final Segmentation and Severity Analysis

Based on the comparison of the three segmentation methods, Adaptive Thresholding was selected as the final method because it achieved the highest Average IoU on the 50 tested images.

The final segmentation result was compared with the original leaf image and the ground truth mask. The predicted mask shows the regions identified as potentially disease-affected.

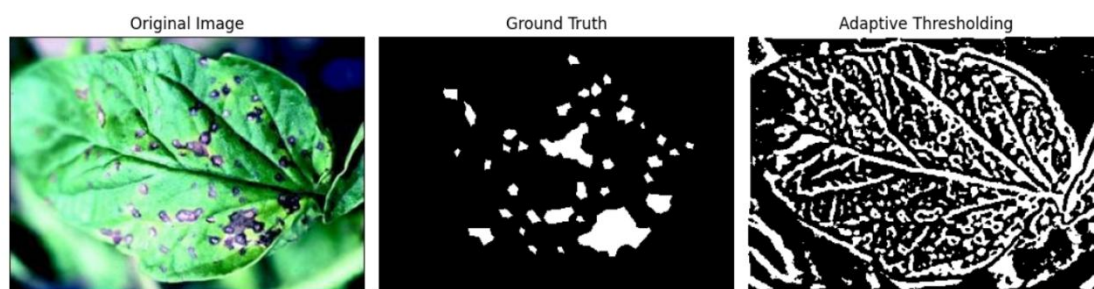
The affected area was estimated by calculating the percentage of pixels identified in the predicted mask. A simple severity level was assigned based on the affected area:

- Less than 10% — Low
- 10% to less than 30% — Moderate
- 30% or more — High

For the sample image used in the final analysis:

- Ground Truth Affected Area: **6.86%**
- Predicted Affected Area: **33.57%**
- Estimated Severity: **High**

The predicted affected area is higher than the ground truth area in this sample. This indicates that the method has over-segmented some regions of the image.



```
▶ # Affected area and severity using Adaptive Thresholding

best_mask = (adaptive_mask > 0).astype(np.uint8)

total_pixels = best_mask.size
affected_pixels = np.sum(best_mask)

affected_percentage = (affected_pixels / total_pixels) * 100

if affected_percentage < 10:
    severity = "Low"
elif affected_percentage < 30:
    severity = "Moderate"
else:
    severity = "High"

print("Estimated Affected Area:", round(affected_percentage, 2), "%")
print("Estimated Severity:", severity)
```

```
... Estimated Affected Area: 33.57 %
    Estimated Severity: High
```

```
# Compare affected area

gt_area = (ground_truth > 0).mean() * 100
pred_area = (adaptive_mask > 0).mean() * 100

print("Ground Truth Affected Area:", round(gt_area, 2), "%")
print("Predicted Affected Area:", round(pred_area, 2), "%")
```

```
Ground Truth Affected Area: 6.86 %
Predicted Affected Area: 33.57 %
```

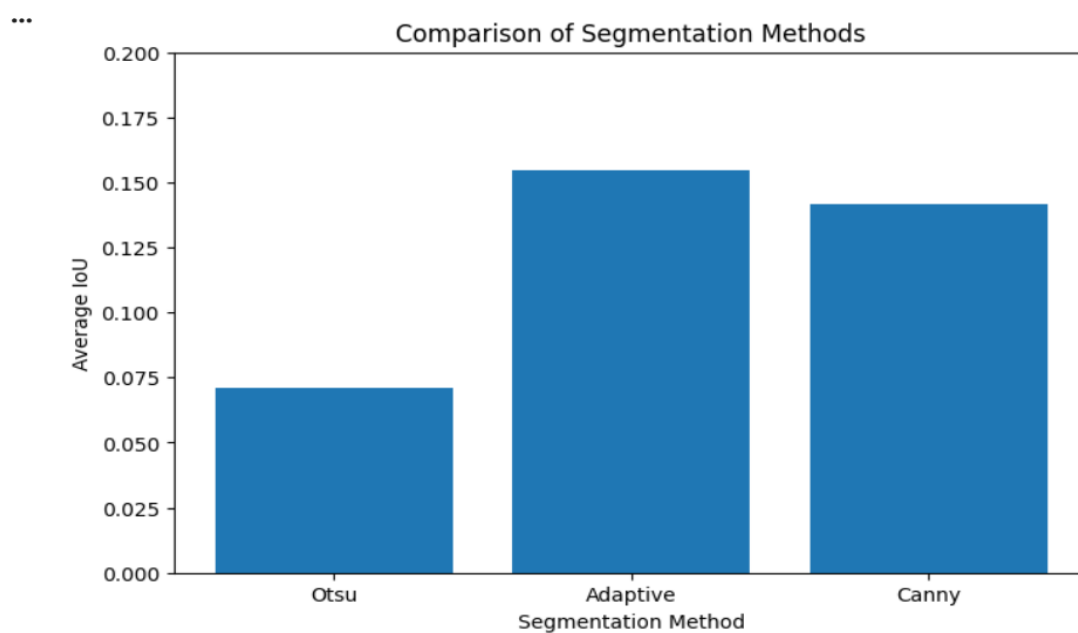
5.8 Visualisation of Affected Region

The final segmentation result is also visualised by highlighting the regions detected as disease-affected. This makes it easier to understand which parts of the leaf were identified by the segmentation method.



5.9 Comparison of Segmentation Methods

The performance of the three segmentation methods is visually compared using their Average IoU values. The graph shows that Adaptive Thresholding achieved the highest Average IoU among the methods tested.



6. Implementation

The project was implemented using **Python in Google Colab**. OpenCV was used for image processing operations, NumPy was used for numerical calculations, and Matplotlib was used for displaying the images and results.

The implementation followed these steps:

1. The dataset was extracted and the image and mask folders were loaded.
2. The input leaf images were converted from colour to grayscale.
3. Gaussian Blur was applied to reduce noise and smooth the images.
4. Otsu Thresholding was applied for segmentation.
5. Morphological opening and closing were used to improve the Otsu segmentation mask.
6. Adaptive Thresholding was applied to handle local intensity variations.
7. Canny Edge Detection was used as an edge-based segmentation method.
8. The three methods were compared using IoU.
9. Adaptive Thresholding was selected based on the highest Average IoU obtained from 50 images.
10. Dice Coefficient and Accuracy were calculated for the selected method.
11. The affected area was estimated from the final predicted mask.
12. A simple severity level was assigned based on the estimated affected area.

7. Results and Discussion

7.1 Segmentation Comparison

The three segmentation methods were tested and compared using the Average IoU obtained from 50 leaf images.

Method	Average IoU
Otsu Thresholding	0.0711
Adaptive Thresholding	0.1547
Canny Edge Detection	0.1419

Adaptive Thresholding achieved the highest Average IoU of **0.1547**. Therefore, it was selected as the final segmentation method.

7.2 Evaluation of Adaptive Thresholding

The selected Adaptive Thresholding method was further evaluated using Dice Coefficient and Accuracy.

- Average Dice Coefficient: 0.2565
- Average Accuracy: 0.7388

These values indicate that the method was able to detect some of the disease-affected regions, but the segmentation was not completely accurate.

7.3 Affected Area and Severity

For the sample image, the ground truth affected area was **6.86%**, while the predicted affected area was **33.57%**. Based on the project threshold, the predicted result was classified as **High severity**.

The difference between the ground truth and predicted area shows that the method can over-segment the affected region.

8. Limitations

The project has some limitations:

- Adaptive Thresholding can sometimes identify healthy regions as diseased, causing over-segmentation.
- The predicted affected area may be higher than the actual affected area.
- The performance of segmentation can change depending on lighting, background, leaf colour, and disease patterns.
- The simple severity calculation is based only on the percentage of pixels in the predicted mask.
- The project focuses on segmentation and does not identify or classify the specific type of plant disease.
- The tested image processing techniques may not perform equally well on every type of leaf image.

9. Future Scope

The project can be improved in several ways in the future:

- More advanced segmentation techniques can be used to improve the accuracy of disease-region detection.
- The system can be tested with a larger and more diverse dataset.
- Image enhancement techniques can be added to handle different lighting and background conditions.
- The severity estimation can be improved using more reliable measurements of the affected region.
- The system can be extended to identify different types of plant diseases using machine learning or deep learning.
- A simple web or mobile application can be developed so that users can upload a leaf image and view the detected affected regions.

10. Conclusion

This project demonstrated the use of Digital Image Processing techniques for segmenting disease-affected regions in plant leaf images.

Three methods — Otsu Thresholding, Adaptive Thresholding, and Canny Edge Detection — were implemented and compared. Based on the Average IoU obtained from 50 images, **Adaptive Thresholding performed the best**, with an Average IoU of **0.1547**.

The project also estimated the affected area of a sample leaf and provided a simple severity indication. Although the method was able to identify possible affected regions, some healthy areas were also detected, showing the limitations of basic segmentation techniques.

Overall, the project shows that Digital Image Processing can be useful for analysing plant leaf images and identifying possible disease-affected regions. Further improvements can make the system more accurate and reliable.

11. GitHub Repository

The complete project source code, dataset-related files, screenshots, and project documentation are available in the GitHub repository.

GitHub Repository:

<https://github.com/mekhajaimon/plant-leaf-disease-segmentation-dip>

